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Unit 1.

Project Preparation

- | 1.1. Project Planification
- | 1.2. Searching for Project Ideas

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1.1. Project Planification UNIT 01

What is data project planification?

- | Series of preparation tasks before starting a data centric project

Ingredients of a successful project planification

- | "Question first approach":
 - ▶ A well defined problem should be the priority.
 - ▶ Considerations about data, method, etc. come later.
- | Focus on the key issues:
 - ▶ Look for key issues first and then expand.

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Project Planification

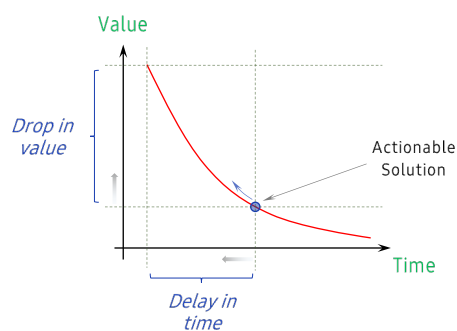
Four project types:

		What to do?	
How to do?	Known	Known	UnKnown
	UnKnown	Optimization	Insight
		Solution	Discovery

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Project Planification

Value vs. Time:



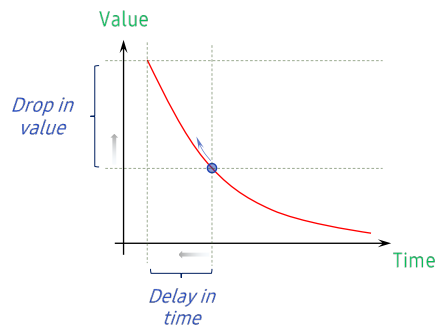
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1.1. Project Planification

UNIT 01

Project Planification

Value vs. Time:



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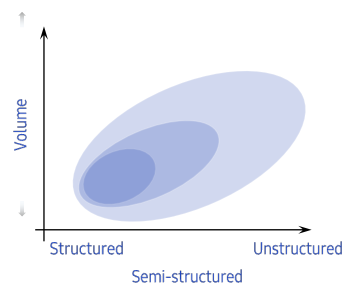
1.1. Project Planification

UNIT 01

Project Planification

Considerations at the planification phase

1) How to acquire the data and which method to use:



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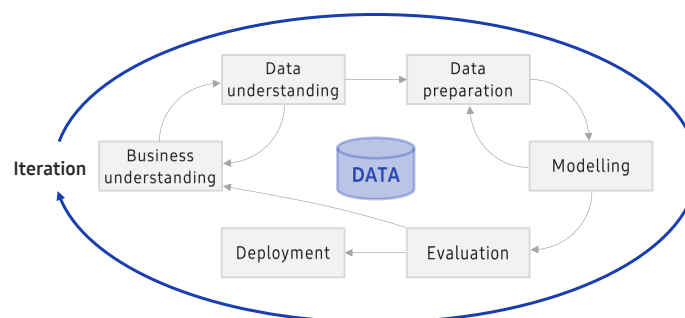
Project Planification

Considerations at the planification phase

- 2) Explore the use cases and preexisting solutions.
- 3) Identify potential hurdles and plan ahead.
- 4) Management of updates and actualizations.

Project Planification

CRISP-DM (Cross Industry Standard Process for Data Mining)



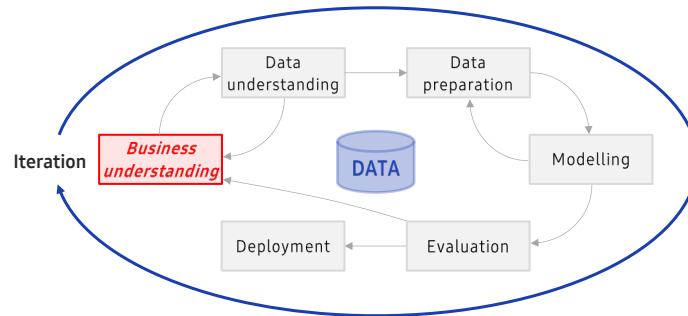
- Structured approach to planning a data mining project

1.1. Project Planification

UNIT 01

Project Planification

| CRISP-DM (Cross Industry Standard Process for Data Mining)



- *Business understanding*: analysis of business objectives and needs

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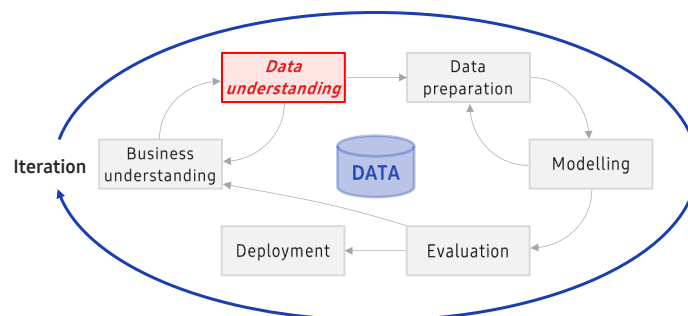
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1.1. Project Planification

UNIT 01

Project Planification

| CRISP-DM (Cross Industry Standard Process for Data Mining)



- *Data understanding*: exploratory analysis to gain further understanding of the gathered data

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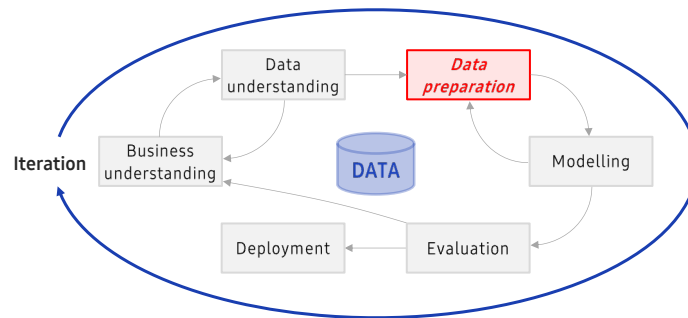
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1.1. Project Planification

UNIT 01

Project Planification

| CRISP-DM (Cross Industry Standard Process for Data Mining)



- *Data preparation*: cleaning, formatting, etc. of the data using the insight gained from the previous steps

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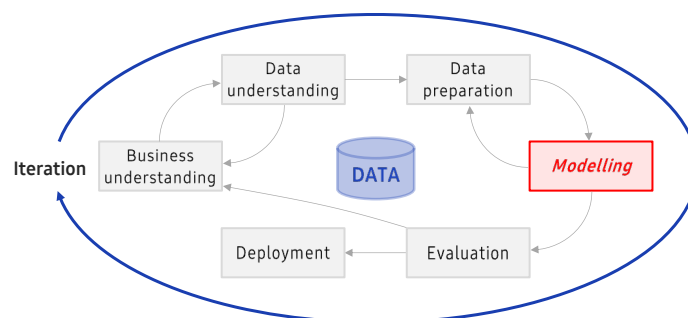
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1.1. Project Planification

UNIT 01

Project Planification

| CRISP-DM (Cross Industry Standard Process for Data Mining)



- *Modelling*: involves selection, optimization and streamlining a predictive model

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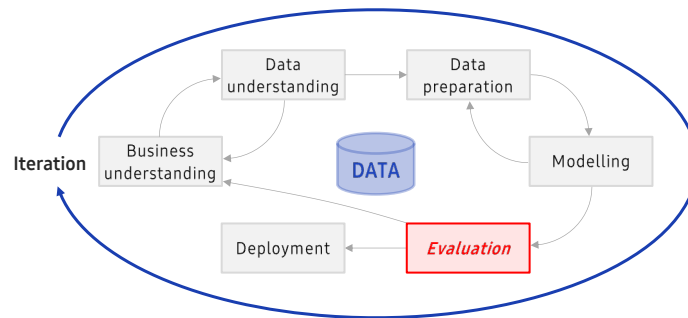
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1.1. Project Planification

UNIT 01

Project Planification

CRISP-DM (Cross Industry Standard Process for Data Mining)



- **Evaluation:** the model from the previous step is put to test in a realistic scenario

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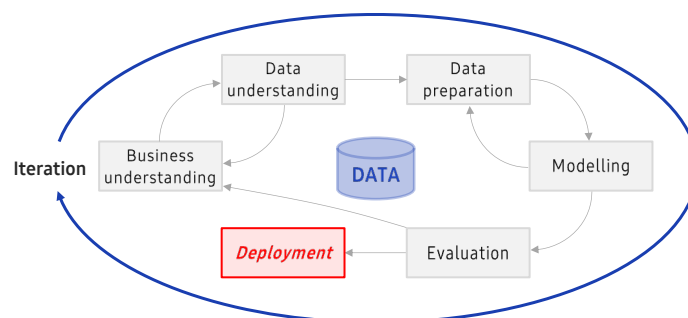
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1.1. Project Planification

UNIT 01

Project Planification

CRISP-DM (Cross Industry Standard Process for Data Mining)



- **Deployment:** the system implemented in a usable format conforming with the business requirements

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Unit 1.

Project Preparation

| 1.1. Project Planification

| 1.2. Searching for Project Ideas

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1.2. Searching for Project Ideas

UNIT 01

Searching for Project Ideas

| Top-Down approach vs. Bottom-Up approach

▶ When the problem is already well defined ⇒ Top-Down approach

▶ The problem needs to be defined through exploration of data ⇒ Bottom-Up approach

	Known	UnKnown
Known	Optimization	Insight
UnKnown	Solution	Discovery

Top-Down Approach

Bottom-Up Approach

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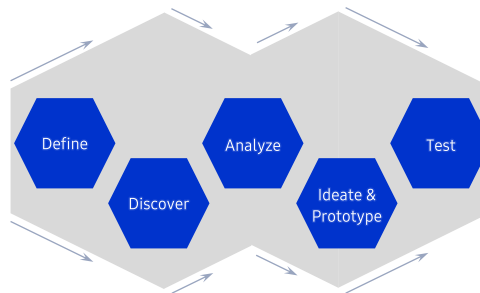
1.2. Searching for Project Ideas

UNIT 01

Searching for Project Ideas

Top-Down approach - Design Thinking

- Five steps: Define, Discover, Analyze, Ideate & Prototype, and Test
- Iterations of convergence and divergence



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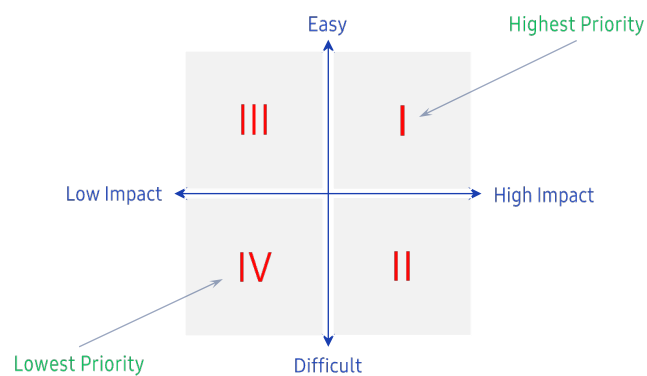
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1.2. Searching for Project Ideas

UNIT 01

Searching for Project Ideas

Top-Down approach - Prioritization



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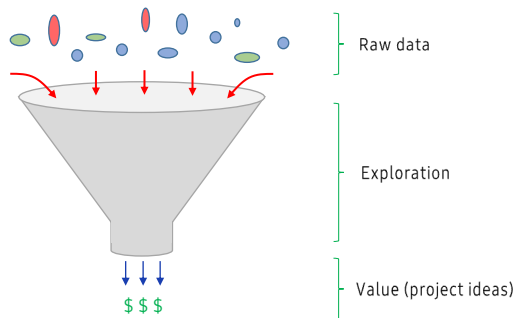
1.2. Searching for Project Ideas

UNIT 01

Searching for Project Ideas

| Bottom-Up approach - Data Exploration

- Explore the raw data to find valuable problems and project ideas.



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Unit 2.

Design Thinking

| 2.1. What is Design Thinking and Why?

| 2.2. Five Steps of Design Thinking

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2.1. What is Design Thinking and Why? UNIT 02

Design Thinking

What is Design Thinking?

- It is a way of thinking that helps you to innovate and solve problems like a designer.



“Design thinking is a human-centered approach to innovation that draws from the designer's toolkit to integrate the needs of people, the possibilities of technology, and the requirements for business success.”

— Tim Brown, CEO of IDEO

Source: ideo.org (<https://www.ideo.org/bio/tim-brown>)

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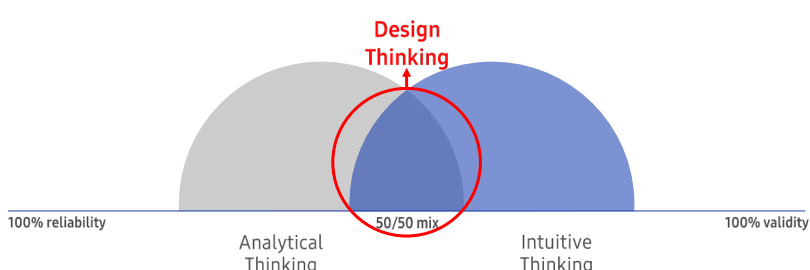
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2.1. What is Design Thinking and Why? UNIT 02

Design Thinking

What is Design Thinking?

BUSINESS THINKING vs. INTUITIVE THINKING



The diagram illustrates the spectrum of thinking from 100% reliability (Analytical Thinking) to 100% validity (Intuitive Thinking). A red circle highlights the intersection of the two, labeled "Design Thinking" and "50/50 mix".

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2.1. What is Design Thinking and Why? UNIT 02

Design Thinking

What is Design Thinking?

KEY ELEMENTS

People-centered	Highly Creative
- Start from what people need or want to do. Empathy is key! - You need the ability to understand and share the feelings of others.	- Design thinking stimulates you to look at situations differently and come up with new solutions that go beyond and improve existing alternatives. Integrative thinking is key. You need the ability to look at all the different aspects of the problem.
Hands-on	Iterative
- Stop discussing, and start working. Make ideas tangible. - Prototyping is thinking with your hands and Test your hypothesis - Failure is a necessary part of the process. Experiments with trial and error are key.	The road to success does not follow a straight line. The more you are able to loop through the "Understand > Create > Learn" cycle, the higher chance you have for good results.

Source: The role of design thinking by Pieter Baert

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2.1. What is Design Thinking and Why? UNIT 02

Design Thinking

What is Design Thinking?

HOW DESIGN FEELS

Source: Designing Better (<http://designingbetter.ca/>)

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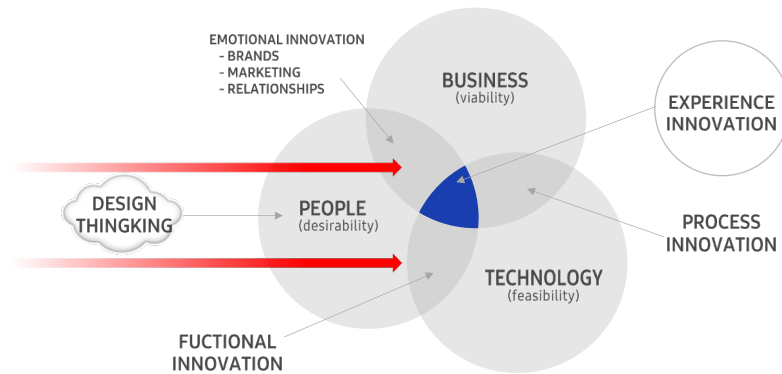
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2.1. What is Design Thinking and Why?

UNIT 02

Design Thinking

What is Design Thinking?



Source: IDEO

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2.1. What is Design Thinking and Why?

UNIT 02

Design Thinking

Design Thinking vs. Analytical Thinking comparison

	Design Thinking	Analytical Thinking
Foundation	Story	Data
Outcome	Insight	Information
Direction	Innovation	Improvement
Starting Point	Empathy	Hypothesis
Methodology	Iteration	Optimization
Procedure	Action first	Plan first
Key value	Human	Physical world

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2.1. What is Design Thinking and Why?

UNIT 02

Design Thinking

Concept of Design Thinking

- ▶ Innovative methodology that interprets a problem in a human-centered way through empathic observations of people and things, clearly defines the problem, then visualizes ideas, rapidly prototypes, and performs tests repeatedly.
- ▶ Design Thinking is a clear and reliable way of solving problems that anyone can use to achieve concrete, practical results.

(Case of application)

- Solve problems with dense and crowded living spaces and minimize stress.
- Use resources around us efficiently and minimize energy consumption.

2.1. What is Design Thinking and Why?

UNIT 02

Design Thinking

Reason for Design Thinking

- ▶ To solve problems related to the people of society.
- ▶ Design thinking is necessary to solve people's problems (the difficulties we face in our daily lives).
- ▶ For example, crowded train stations, online banking with poor Internet transmission, and traffic congestion are occurring in our daily lives, and we are facing frustration and social problems.
- ▶ By applying the principle of design thinking, rational and the best possible alternatives can solve these problems better, faster, and smoothly.

Types of problems that can be resolved by Design Thinking

- ▶ Issues not clearly defined
- ▶ Problems Insufficient Information to Create a Solution
- ▶ Problems that Strongly Affect People's Needs
- ▶ Problems with changing circumstances

2.1. What is Design Thinking and Why?

UNIT 02

Design Thinking

Case Study - Embrace Infant Warmer

- ▶ Start with a design school project by four Stanford University students
- ▶ 4 million babies die within the first month after birth
- ▶ Accurate Understanding of Incubators: Optimal Products, Not Low-cost Products
- ▶ Newborn warmers priced at 1% of existing incubators



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2.1. What is Design Thinking and Why?

UNIT 02

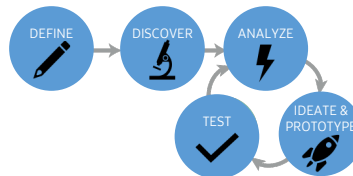
Design Thinking

Four Characteristics of Design Thinking

- ▶ 1) Planning the design is not "brainstorming," nor is it a "one-day work process."

The design execution plan is not an instant process. It takes the time and effort of dedicated and committed members to participate in achieving a common purpose.

- ▶ 2) Work repeatedly in design thinking.



Planning the design is an iterative process, and each step needs to be repeated in the forward and backward directions. This iteration makes a variety of discoveries and ultimately arrives at the desired result acquisition or decision point.

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2.1. What is Design Thinking and Why?

UNIT 02

Design Thinking

Four Characteristics of Design Thinking

- ▶ 3) In the design action plan, all participants should be able to collaborate and work together.

In a true Design Thinking process, every participant (investigators, business leaders, designers, engineers, etc.) must be involved in everything from start to finish.

All team members should work together to gain deep insights (a deep understanding of what the real problem is and what users or customers are asking for).

- ▶ 4) Each step of planning the design has an output.

As each step of Design Thinking is carried out, various outputs that can be used as evaluation criteria are produced. Each deliverable is not required and not all deliverables. Optionally create the required deliverables, given the nature of the projects involved.

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2.1. What is Design Thinking and Why?

UNIT 02

Design Thinking

Required list of materials for Design Thinking

- ▶ Suitable collaboration space
- ▶ Paper
- ▶ Writing instruments
- ▶ Tools for Cutting or Pasting
- ▶ Camera / Images
- ▶ Sketchbook
- ▶ Post-it
- ▶ Cardboard
- ▶ Whiteboard / Wall / Pinboard
- ▶ Card
- ▶ Prototyping Materials
- ▶ Framework

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Unit 2. Design Thinking

| 2.1. What is Design Thinking and Why?

| 2.2. Five Steps of Design Thinking

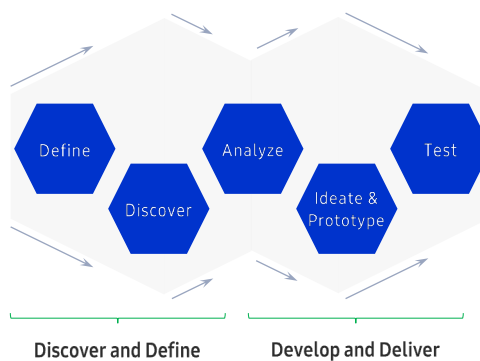
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2.2. Five Steps of Design Thinking

UNIT 02

Design Thinking

| Convergence and Divergence



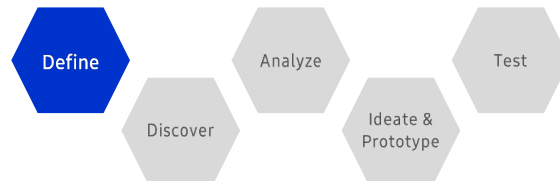
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2.2. Five Steps of Design Thinking

UNIT 02

Define

Step #1: Define



- ▶ Use the insight obtained from observing and talking.
- ▶ Identify problems that are highly **valuable**, **inspiring**, and **real**.
- ▶ Answers to "who", "what", "why", etc.

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2.2. Five Steps of Design Thinking

UNIT 02

Define

Guide to writing problem definition proposals:

- ▶ 1. What problem do you want to solve? Please input in the following sentence:

What is a method of [Where] [Who] [What problem] [-Do not ~] [-Do ~]

- ▶ e.g. 1. [Children's House] [Nursery teachers] [In a situation where parents' distrust prevails] [without misunderstanding] [What is the way to communicate with their parents?] → Preparing tools for communication between nursery teachers and parents
- ▶ e.g. 2. [Over the world] [Disabled] [In a situation where you can't communicate with others] [without expensive money] [What is the method for social networking?] → Communication device design for the disabled by using proper technology
- ▶ e.g. 3. [Every day at school] [Students] [when they eat] [without compulsory means] [How can they reduce the amount of food waste?] → Design the solution for the remnants of school meals

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2.2. Five Steps of Design Thinking

UNIT 02

Define

Guide to writing problem definition proposals:

- ▶ 2. How did you come across this problem? And why do you want to solve it?
- ▶ 3. What do you think is the cause of this problem?
- ▶ 4. How will this work out for the world? Explain how the effects can be measured.
 - ▶ If this problem is solved,
 - 1) Social effects: through (example)
 - 2) Personal effects: through (example)

2.2. Five Steps of Design Thinking

UNIT 02

Define

Guide to writing problem definition proposals (Example):

- ▶ 1. What problem do you want to solve? Please input in the following sentence.
What is a method of [Where] [Who] [What problem] [-Do not ~] [-Do ~]
 - ▶ **What is the way to [diagnose ocular disease] [of people with low income] [in developing countries] [to provide timely treatment to prevent avoidable blindness and visual impairment]?**
- ▶ 2. How did you come across this problem? And why do you want to solve it?
 - ▶ **Of 285 million people who are visually impaired, 90% are living in developing countries. Over 90% of visual impairment result from chronic causes, and over 80% of visual impairment could have been prevented by a proper intervention at each life stage.**
 - ▶ **Underprivileged people are more prone to contract a disease than other people. Visual impairment is one of the main causes of poverty, and once those who are visually impaired recover, they can return to an economically active population and recover productivity.**

2.2. Five Steps of Design Thinking

UNIT 02

Define

Guide to writing problem definition proposals (Example):

- ▶ 3. What do you think is the cause of this problem?
 - ▶ **Most ophthalmoscopes are not affordable to be implemented locally in developing countries. Thus, early detection of ocular diseases whose timely diagnosis is critical to treatment is extremely unachievable.**
- ▶ 4. How will this work out for the world? Explain how the effects can be measured.
 - ▶ **If this problem is solved,**

Social effects: social costs will be reduced by preventing blindness in developing countries.

Personal effects: through early detection of ocular disease, individuals will suffer less from poverty resulting from decreased productivity due to visual impairment.

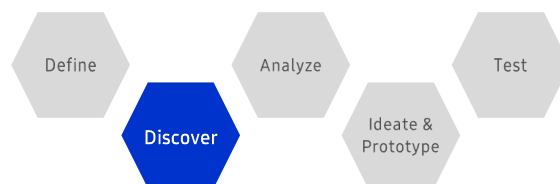
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2.2. Five Steps of Design Thinking

UNIT 02

Discover

Step #2: Discover



- ▶ In this stage, data research, observation research, and interview are primarily done.
- ▶ Observe, talk, and feel!

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2.2. Five Steps of Design Thinking

UNIT 02

Discover

Process

- In the situational discover stage, the first step is to investigate the existing information related to the problem and the solution of others through the 'data survey' and to construct the hypothesis through your own critical thinking.

```

graph LR
    A["Data Research  
Academic Data / Magazine /  
Articles / Statistics"] --> B["Observation Research  
People / Environments /  
Tools / Activities"]
    B --> C["Interview  
Recipient Interview /  
Specialist Interview"]
  
```

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2.2. Five Steps of Design Thinking

UNIT 02

Discover

Data (Document) Research

- Good research is essential to the process of delivering effective solutions.
- When you find a problem you want to solve, you need to know what research has already been done in the relevant field to come up with an influential solution that will help you solve the problem. Rather than answering and approaching problem-solving, it's important to look for clues that can help you find unexpected solutions through the situation observation phase.
- If you're new to the topic, you'll need relevant books, articles, research, papers, and web documents to help you get background and problem information, explore a vast amount of resources, and find effective ways to solve your challenges.

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2.2. Five Steps of Design Thinking

UNIT 02

Discover

| Data (Document) Research

- ▶ 1) Is it reliable?
 - ▶ When looking for material, check whether the author of the material is an expert in the field, if it was created too long ago, and whether the information is from a trusted source or individual. For example, blog posts from non-professionals, intellectual answers from search portals, etc., are information with low credibility.
- ▶ 2) Isn't it duplicated?
 - ▶ Look for items that have been presented in the past with research, best practices, and problem-solving solutions, and make sure that other solutions aren't already solved or ideas that overlap with recent technology advances. Creating a simple timeline can help. Summarize important content in relevant resources and use the team's critical thinking to think about the significance and application of the material.

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2.2. Five Steps of Design Thinking

UNIT 02

Discover

| Observation and Investigation

- ▶ Based on the information obtained through the research, we select a place related to the problem and visit.
- ▶ You can find the right place to experience the situation in question directly or indirectly and then observe the subject in the field and see unconscious and natural behavior that is not expressed in language. The richer the findings and questions from observational investigations, the easier it is to determine where the challenges are to be addressed in the next step, the orientation phase.

01. Think **freely**

- There is no one right answer.
- Put yourself in the shoes of the person experiencing the problem and look at the problem from a new perspective.
- Write freely what they see, what they see, and what they question.
- Even minor findings can lead to very meaningful solutions, so think without limiting them.

02. Looking at a **New Perspective**

- Observe the things that have been taken for granted.
- You may see things you've never seen before and discover new problems.
- For example, are bookstores where people go to buy books?

03. **Record** in Detail

- Writing down findings, questions, etc. in detail develops ideas and leads to more ideas.
- Your records will serve as an important foundation for finding true solutions in the next step, Direction Finder.

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2.2. Five Steps of Design Thinking

UNIT 02

Discover

| Observation and Investigation

- ▶ Visiting the site with team members, the team members observe 15 minutes each with four perspectives (people, tools, environment, and activities) of the survey guide.
- ▶ Organize the observations, findings, and questions related to each view in the appropriate guide plate.

 PEOPLE Observing people	 TOOLS Observing tools
 ENVIRONMENT Observing environment	 ACTIVITY Observing activity

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2.2. Five Steps of Design Thinking

UNIT 02

Discover

| Observation and investigation (Example)

 PEOPLE Observing people	 TOOLS Observing tools
▶ People who do not afford to go to regional medical clinics visit public health centers	▶ Receive eye check ups without fundus camera
 ENVIRONMENT Observing environment	 ACTIVITY Observing activity
▶ Ocular diseases need to be diagnosed early and given proper treatments ▶ Most people in developing countries miss this chance when they visit public health centers	▶ Most people are prescribed simple analgesics without being diagnosed with disease and are dissatisfied with the treatment

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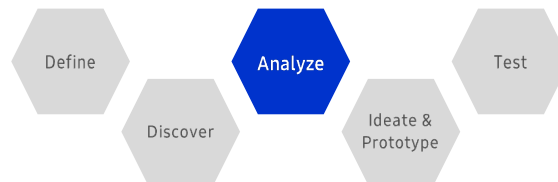
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2.2. Five Steps of Design Thinking

UNIT 02

Analyze

Step #3: Analyze



- ▶ Interiorize the problem as one's own.
- ▶ Learn more about persona and understand situation and circumstances more deeply.

2.2. Five Steps of Design Thinking

UNIT 02

Analyze

Persona

- ▶ Definition
 - ▶ Persona is a hypothetical person who represents the type of people who can use a product or service and is based on data obtained from various surveys.
 - ▶ Through the process of setting up the Persona we think is who the solution is, thinking about the beneficiary's position, and building consensus, the Persona created in the Analyze process continues to be a tool to identify what a recipient really needs. If we can truly understand and empathize with the Persona we've created, the solution we're going to create will be a necessary, sustainable solution for the world.
- ▶ What to consider when creating Personas
 - ▶ Based on the research, we found ourselves to predict most closely how someone will behave in a particular situation and environment. You don't need to use the actual survey, but like introducing my best friend, draw a Persona with a distinct personality.
 - ▶ You don't have to worry that Persona will look biased. Personas don't have to be generalized. They characteristically paint Personas even though they're a bit extreme.

2.2. Five Steps of Design Thinking

UNIT 02

Analyze

Persona

- 1. Demographic
 - By setting objective elements that describe a persona, such as age, gender, and occupation, it is easy for anyone to think about who the person is. By subdividing the persona with this basic information, it becomes clear which group of people the persona represents.

I am _____ years old.

My job is _____, and what I do is _____.

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2.2. Five Steps of Design Thinking

UNIT 02

Analyze

Persona

- 2. Touchpoint
 - What kind of person do we set up to meet people, what things to carry, and in what environment?
 - By looking at Persona's life through social groups, essential items, and key environments, we can map out specifically what the recipient wants in an environment and what factors should be considered for the solution to be meaningful to be used for future development.
 - Social Group: What kind of people does Persona live with, talk to, and share her thoughts with? I think about the people around Persona, and I think about who and why it is now.
 - Essential Items: What items does Persona carry around all the time or are important items? While looking at the essential items, we also think about why such items are essential for Persona.
 - Major Environments: Where do Personas usually spend their time? Think about it all, including physical environments such as schools, stores, banks, and bus stops, as well as digital environments such as websites, SNS pages, and mobile apps.

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2.2. Five Steps of Design Thinking

UNIT 02

Analyze

Persona

3. Value

- ▶ Based on the findings from various surveys, it defines the 'value' that the beneficiary thinks meaningfully. What does this person think as the most important thing? What makes this person happy? Is there anything you try to keep?
- ▶ Based on the following questions, try to find the core value our Persona has.
- ▶ It is very important to find the core value that Persona thinks is meaningful to solve the problem in the right direction.

What amuses me is
 What I think is important to me is
 What I hate very much is
 My goal is

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2.2. Five Steps of Design Thinking

UNIT 02

Analyze

Persona

4. Behavior

- ▶ Describes actions that reflect the core values of Persona.
- ▶ Even in the same problematic situation, people's behavior can vary depending on what they value.
- ▶ Consider and list the behaviors that directly or indirectly relate to the problems we have defined that best represent Persona's thoughts.

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2.2. Five Steps of Design Thinking

UNIT 02

Analyze

Persona

5. Wants

- ▶ Describe what the beneficiary wants based on his or her core values and behavior.
- ▶ By thinking specifically about what the beneficiary wants and expressing it in one sentence, we set the direction of the solution we will develop in the future.
- ▶ What are some inconveniences Persona feels, and in what direction do you want them to be resolved in the future?
- ▶ What is the first thing that needs to be addressed? What can be better now?
- ▶ Look at the wants described in Persona's mind and thought about their priorities.

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2.2. Five Steps of Design Thinking

UNIT 02

Analyze

Persona (Example)

I am 55 years old.
 I am a daily wage construction worker, and what I do is building houses located in the suburban area.
 What amuses me is a meal voucher provided by the municipality.
 What I think is important to me is saving money as much as I can since I might have no work to do if I get older.
 What I hate very much is the fact that blurriness in the eye has disturbed me a lot recently.
 My goal is to achieve economic stability without worrying about subsistence.

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2.2. Five Steps of Design Thinking

UNIT 02

Analyze

Persona (Example):



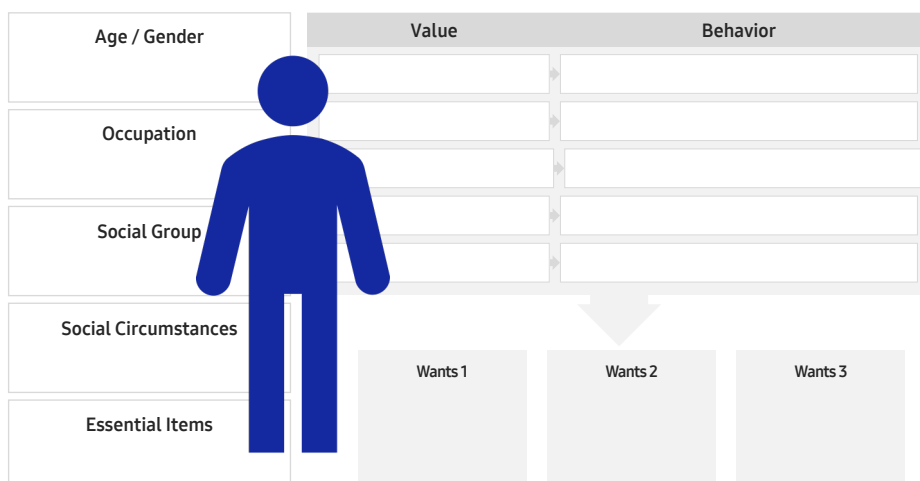
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Appendix. Persona

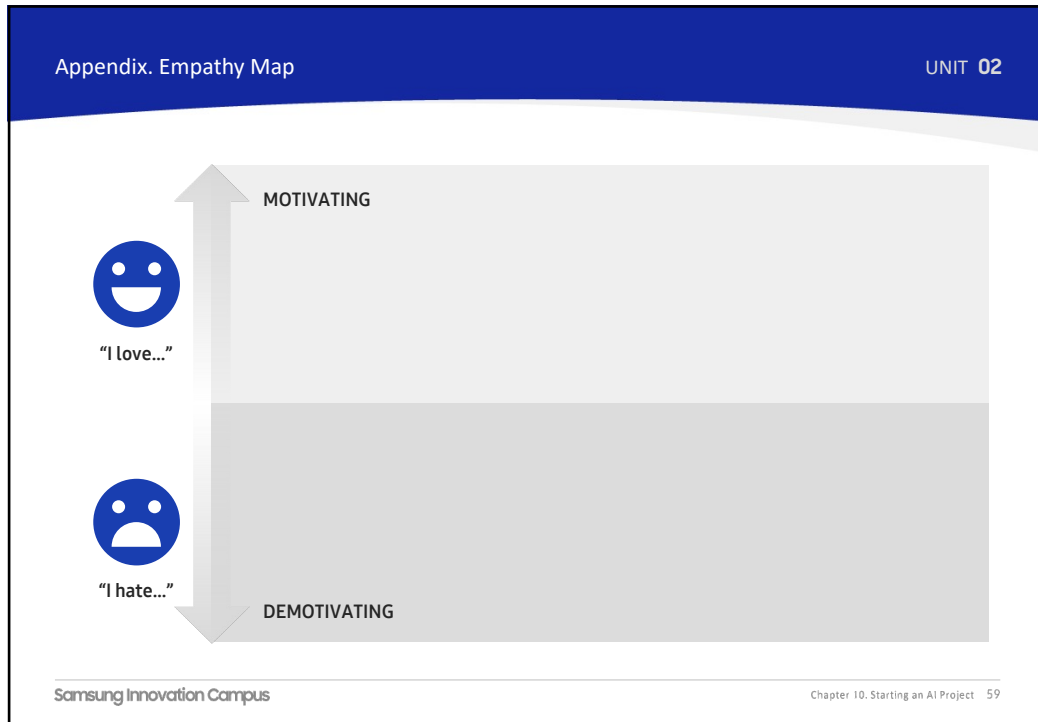
UNIT 02



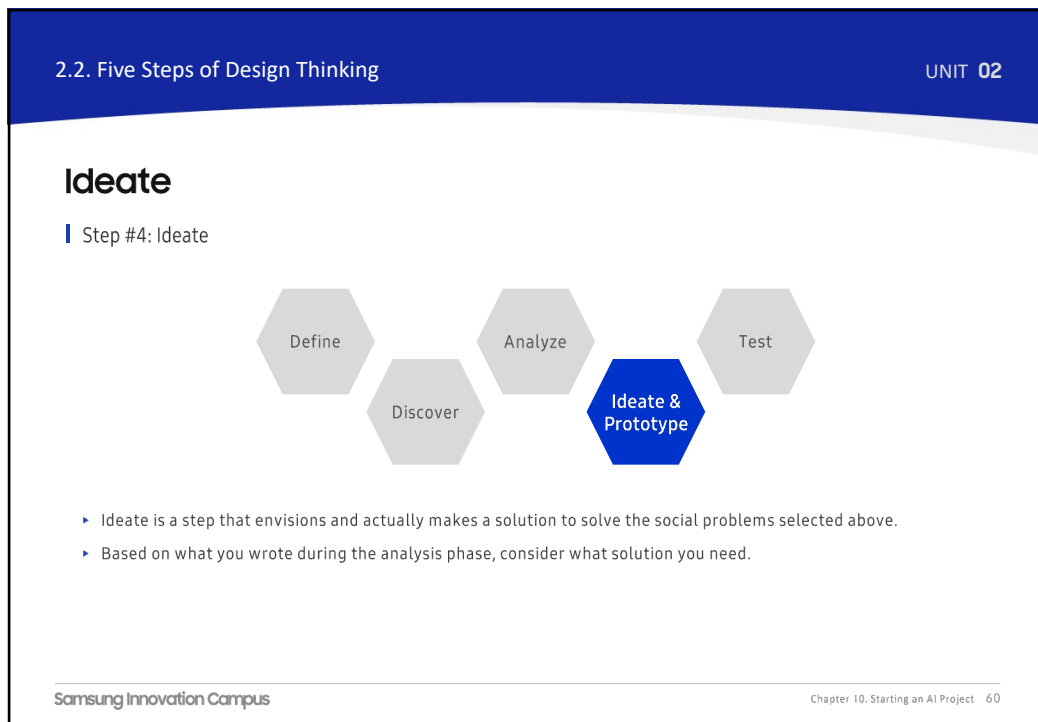
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2.2. Five Steps of Design Thinking

UNIT 02

Ideate

Brainstorming

- ▶ Definition and Necessity of Brainstorming
 - ▶ Brainstorming is a method that puts as many ideas as possible as if a storm is striking the head. As you share ideas with many people, not thinking about them alone, you can use various perspectives and make a better solution by combining ideas.

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2.2. Five Steps of Design Thinking

UNIT 02

Ideate

Brainstorming

- ▶ Things to consider during Brainstorming
 - ▶ Put as many ideas as possible. An idea can give birth to another idea. Brainstorm again with the newborn idea and come up with more ideas.
 - ▶ When putting ideas, use 'Post It.' You can easily share when you write a single idea down on one 'Post It'. The process of grouping and prioritizing can be more convenient, too. Also, if necessary, you can draw a picture to replace a detailed explanation.
 - ▶ Proceed after you decide the time. Without another limit, write ideas down continuously until the decided time.
 - ▶ If the flow of thoughts gets slowed down, check the idea list. Some of the ideas can be better and give birth to another idea.
 - ▶ You don't have to deduce the conclusion while you are brainstorming. Do not be hasty in the ideate stage. Think whether there is a way to make it a better idea.

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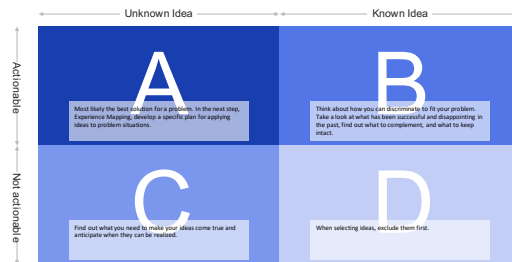
2.2. Five Steps of Design Thinking

UNIT 02

Ideate

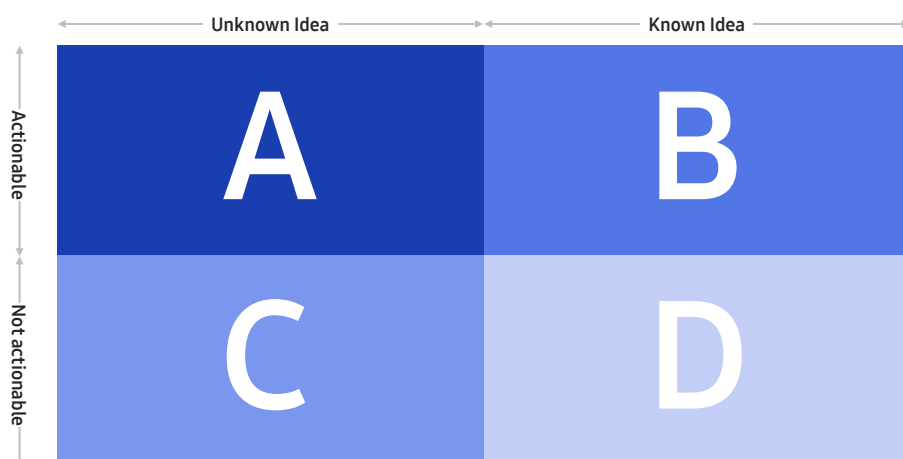
Idea Grouping

- ▶ Through Idea Matrix, make a group from the ideas deduced by brainstorming and prioritize them. Divide into two criteria; creativity(horizontal) and plausibility(vertical). Then you will know what idea to solutionize for the first thing.
- ▶ Discuss together whether it has been tried, whether there is to benchmark, or whether it is already executed. To determine plausibility, think together about what is a required technology or infrastructure and whether there is a better solution in terms of effect and cost. When you choose an idea, it is effective to do it with specialists. Proceed with the specialist matched to your team.



Appendix. 2 by 2 matrix

UNIT 02



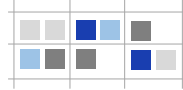
Appendix. Sticky Notes
UNIT 02



Cluster and Piles



Links



Grids



Axes



Circles and Targets



Metaphors

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2.2. Five Steps of Design Thinking
UNIT 02

Ideate

Experience Mapping

- ▶ Definition
 - ▶ Experience Mapping is a method of developing an ideal beneficiary experience. When you apply the solution to the problem, it plans all elements required to provide the most ideal experience. Through mapping, you get to know to whom, when, how, and what benefits the solution provides.
- ▶ Things to consider Experience Mapping
 - ▶ Keep in mind that the subject of Experience Mapping is the Persona. Think about the experience from the beneficiary's perspective.
 - ▶ All solutions are not like a magical thing that happens instantaneously. They are composed of experiences and a chain of moments. Through Experience Mapping, check once again whether you designed the solution after capturing the right moment, place, and person for the solution.

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2.2. Five Steps of Design Thinking

UNIT 02

Ideate

| Experience Mapping

▶ Experience Mapping How-to

- ▶ 1) Decide the whole context of the problematic situation and think about in what order what touchpoints are linked.
- ▶ 2) Put the actions that are expected at each touchpoint.
- ▶ 3) To create a rapport and elevate understanding, visualize the touchpoints and the happening things and explain them in a single story.
- ▶ 4) After looking at the big picture, figure out how the current problem can turn out if the chosen idea becomes a real solution.
 - ▶ Is this really the necessary solution for the beneficiary?
 - ▶ Is the part to which the solution is applied the most suitable touchpoint for the solution?
 - ▶ For the solution to work, what must be done before and after the solution?
- ▶ 5) Find a way that can complement the solution.

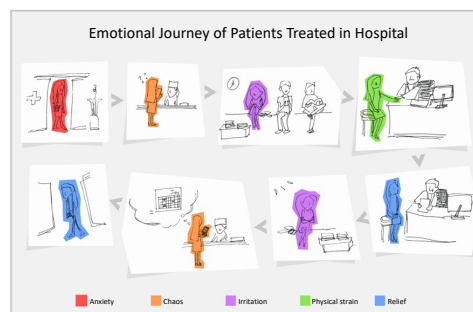
2.2. Five Steps of Design Thinking

UNIT 02

Ideate

| Experience Mapping

- ▶ "Way to express can differ according to on which touchpoints to put the focus on."
 - ▶ Emotional Journey



2.2. Five Steps of Design Thinking

UNIT 02

Ideate

| Prototyping

- ▶ Definition and necessity of Prototyping
 - ▶ A way to improve the solution is through adding, changing, and deleting things to the point of satisfaction according to the real-using evaluation.
 - ▶ This not only makes the optimal solution through repetitive trying and complementation but also helps to consider the beneficiary's perspective. Hence, it helps a lot in thinking, too.
- ▶ Things to consider for Prototyping
 - ▶ Prototyping is a starting point that materializes ideas that exist only in mind. Make and fail, re-make and improve it, and then complete the best solution. Not any idea can be perfect from the start. Do not be afraid of failure. 'Start' and 'Repeat.'
 - ▶ Improve the solution through repetition. It does not mean pure repetition. You have to reflect more precisely on the beneficiary's needs at every interval.

2.2. Five Steps of Design Thinking

UNIT 02

Ideate

| Prototyping

- ▶ Prototyping Tool: Products (for reference)
 - ▶ For products, it is the most certain way to try various forms and complement them quickly.
 - ▶ To check various functions, location of the operation button, and ergonomic convenience, it helps to make a form with simple materials such as paper, clay, Styrofoam, and acryl board. To implement a certain design, try designing with the 3D Modelling Program.



2.2. Five Steps of Design Thinking

UNIT 02

Ideate

Prototyping

- Prototyping Tool: UI (for reference)
 - As things are digitalized, various services that help make mobile applications and websites are coming out. Refer to the site below and try to make a wireframe and architecture.
 - Reference:
 - Flinto (<https://www.flinto.com/>)
 - Balsamiq (<https://balsamiq.com>)
 - Balsamiq introduction video (<https://www.youtube.com/watch?v=zLysy3IPfFI>)
 - Paper prototype: Homepage making (<https://www.youtube.com/watch?v=ppnRQD06ggY#t=228>)

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2.2. Five Steps of Design Thinking

UNIT 02

Ideate

Prototyping

- Prototyping Tool: Service (for reference)
 - Research resources are generally very useful for gathering the necessary information, but it is difficult to show the full context or tell the story. Scenarios are a way to compensate for this and provide a realistic picture of the problem situation and create an expected situation to clearly communicate the effect when the solution is applied. In particular, the service solution is difficult to have a physical form or platform, such as a product or UI, so you can try how the service proceeds or interact with the beneficiaries through scenarios.
 - 1) Design the whole story

Recall the series of moments your beneficiary experiences in a problem with the solution. The scenes should be organized in a structured way around the differences, including how existing problems are solved and what the beneficiaries will benefit from.

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2.2. Five Steps of Design Thinking

UNIT 02

Ideate

| Prototyping

- ▶ Prototyping Tool: Service (for reference)

- ▶ 2) Decide how to express

Think about how you can tell the story effectively. If necessary, it can be expressed in various ways such as concept board and shooting video.

- ▶ 3) Refine your solution

Think about what you need for each step to run the service. Think about what guides are needed for beneficiaries who use the services and if there are no protocols for service providers to follow.

2.2. Five Steps of Design Thinking

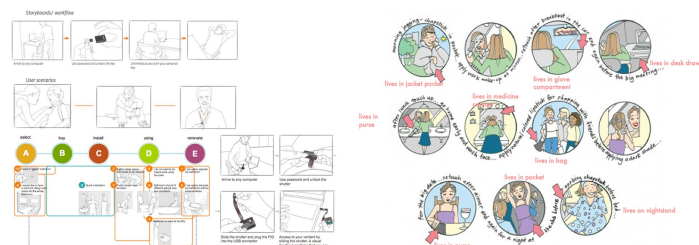
UNIT 02

Ideate

| Prototyping

- ▶ Prototyping Tool: Service (for reference)

- ▶ Concept boards are the most commonly used method, which allows you to represent what happens when a solution is applied, step by step, according to changes in action or location.



2.2. Five Steps of Design Thinking

UNIT 02

Test

Step #5: Test

```

graph LR
    Define[Define] --- Discover[Discover]
    Discover --- Analyze[Analyze]
    Analyze --- Ideate[Ideate & Prototype]
    Ideate --- Test[Test]
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    style Define fill:#d9d9d9
    style Discover fill:#d9d9d9
    style Analyze fill:#d9d9d9
    style Ideate fill:#d9d9d9
  
```

- ▶ Test the prototype in a realistic scenario.
- ▶ Be critical. Avoid being defensive.
- ▶ Find the deficiencies and improve!

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2.2. Five Steps of Design Thinking

UNIT 02

Test

Definition

- ▶ Definition and necessity of Test
 - ▶ It is easy to misunderstand the Test as a gateway that determines success and failure.
 - ▶ However, the Test is the last stage of completing the solution. It inspects the prototyped solution in a real situation of the problem and finds room to improve and complement it.
 - ▶ If you apply the solution to the beneficiaries, you can check whether it is necessary.

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2.2. Five Steps of Design Thinking

UNIT 02

Test

Pre-Test: Test Preparation

- ▶ 1) WHERE: Find a place and environment to apply the solution.
 - ▶ Where is the place where the test is operated?
 - ▶ Did you get permission from the facility manager?
- ▶ 2) WHO: Find the same beneficiary as the previously stated Persona.

Solutions based on Persona, as itself, represent something. The effect of the solution gets more noticeable when the object of the test is a beneficiary in the situation. Find the beneficiary who can participate over the things written in Persona such as age, job, social group, social circumstances, essential item, etc. Even if it is the same Persona, you'd better test about 5 people not to lean toward one person's opinion.

- ▶ Who are the participants of the test? Are they participants based on Persona?
- ▶ How many are they?

2.2. Five Steps of Design Thinking

UNIT 02

Test

Pre-Test: Test Preparation

- ▶ 3) WHAT: Establishes evaluation criteria that consider both functional and emotional aspects.

To assess the effectiveness of your solution, see whether it solves the problem functionally or satisfies the emotions you need.

- ▶ Is it an evaluation standard that considers both functional and emotional aspects?
- ▶ Did the criteria reflect what you wanted to solve with our solution?
- ▶ Can the metrics measure the impact before and after the solution?

Functions	Emotions
• How convenient is it? • How efficient is it? • Is it a system that anyone can learn easily? • Is it designed for the beneficiary's behavior? • Is it causing another problem? (Efficiency and convenience)	• What is the beneficiary's satisfaction with problem-solving? • Does it match the lifestyle of the beneficiary? • What impact does this change have on the beneficiary's life? • Does it cause another problem? (Emotional stability)

2.2. Five Steps of Design Thinking

UNIT 02

Test

Pre-Test: Test Preparation

- ▶ 4) HOW: Plan effective test methods.

Think about how to test your solution. Make practical plans such as the schedule, audience, and location, as well as how you will observe the beneficiary, what will be interviewed, and how many people will be present.

- ▶ Are the schedule, destination, location, etc. confirmed?
- ▶ Are the roles of the team members interviewed clearly defined?
- ▶ What would we like to test?

2.2. Five Steps of Design Thinking

UNIT 02

Test

Test: Test Execution

- ▶ 1) EMPATHY: Be a beneficiary.
 - ▶ IS WHAT: Thinking about the persona profile you created earlier, this is a way to experience the problem situation where the solution is applied directly like a beneficiary.
 - ▶ DO WHAT: Predict what behavior your beneficiaries will see when experiencing a new system they are unfamiliar with, and see if they can easily accept the solution and see if there are any problems.
 - ▶ TIP: It's a good idea to run this method as a means of checking the solution one more time before starting a full-scale test on beneficiaries.

2.2. Five Steps of Design Thinking

UNIT 02

Test

| Test: Test Execution

- ▶ 2) OBSERVATION: Observe the beneficiary.
 - ▶ IS WHAT: Applying the solution to the actual situation and observing the beneficiary's response under the same conditions as usual.
 - ▶ DO WHAT: Write down what you want to see and observe the site in advance, as expected, whether the solution works and what happens to the beneficiary's behavior. Evaluate each item to see what and how to improve it.
 - ▶ TIP: Observe the solution's effectiveness by observing the beneficiary's most natural behavior in the same way as the observation that you did during the Discover phase.

2.2. Five Steps of Design Thinking

UNIT 02

Test

| Test: Test Execution

- ▶ 3) INTERVIEW: Ask the beneficiary.
 - ▶ IS WHAT: Interview the beneficiary about what they need to know about the solution. Most commonly, the beneficiary first experiences the environment in which the solution is applied and then asks questions about it.
 - ▶ DO WHAT: The interview asks the beneficiaries for their opinions and advice on what they like, what is uncomfortable or unnecessary, and what needs to be compensated for.
 - ▶ TIP: The beneficiary may be able to ask questions and experience depending on the solution type and may be interviewed after a week of testing.
 - ▶ ✳ When conducting an interview after the test period, you must pre-select the test period and provide a mission or assignment that the beneficiary must complete to determine what they want to know during the test period.

2.2. Five Steps of Design Thinking

UNIT 02

Test

Post-Test: Test Results Summary

- ▶ 1) REACTION & RESULT: Measure the beneficiary's response and the solution's effectiveness.
 - ▶ See how the beneficiary responds, how the solution actually solves the problem, how satisfied the beneficiaries are, and whether there are any emerging problems or unexpected effects in applying the solution.
- ▶ 2) NEXT STEP: Find the missing parts of the solution and find ways to fix it.
 - ▶ Apply the solution in real life, and think about how to modify and supplement it. Find out what you need to do to increase the effectiveness of your problem-solving and plan a specific action plan. And validate...
 - ▶ Business plan that describes how to commercialize our solution in the future
 - ▶ Selected technology partners and experts who think what our solution needs
 - ▶ How much money will you need, and how will you get support?
 - ▶ Know when it's feasible and what's the specific action plan for the future



